Generalized TD/GAE Estimator Perturbation: Bias, Stationarity, and Training-Time Policy Manipulation

Abstract

We study how a perturbation added to the computed advantage estimator changes the policy-gradient update. The main distinction is between total estimator bias and its action-dependent component: a bias shared by all actions at a state has no direct same-step score contribution, whereas within-state variation produces a gradient perturbation. Under standard smoothness, bounded-score, and stochastic-noise assumptions, we prove a finite-time weighted-average stationarity bound for the resulting biased ascent method. Asymmetric TD-error clipping is retained as a special case of the general perturbation, while the main experiments use a calibrated, explicitly action-selective advantage-level perturbation.

The security claim is deliberately constrained. If an attacker can rewrite the entire training stack, including weights, gradients, optimizer state, and telemetry, an estimator-level claim is not meaningful. We instead study a compromised advantage-producing component with no direct access to environment data, sampled actions, policy parameters, optimizer state, critic targets, or deployment code. The practical question is whether a small targeted change to the advantage tensor can alter the policy through ordinary PPO updates, while conventional PPO telemetry remains difficult to distinguish during training. The experiments separate estimator diagnostics from online training and end with a frozen-policy action and environment-level test.

1 Problem and Threat Model

Reinforcement-learning poisoning is often summarized by a drop in final return or by attack success. Those outcomes do not identify the mechanism that turns a small training-time manipulation into a persistent policy change. We analyze that mechanism for an on-policy actor–critic learner.

Consider a distributed PPO system in which simulator or robot workers upload clean transitions

$$(s_t, a_t, r_t, s_{t+1}, d_t),$$

where d_t indicates a terminal transition. The clean learner first forms

$$\delta_t = r_t + \gamma(1 - d_t)V_\omega(s_{t+1}) - V_\omega(s_t), \qquad \widehat{A}_t = \mathrm{GAE}(\delta_{t:T}).$$

In the primary threat model, the attacker compromises the component that hands advantages to the actor update. It receives the clean advantage tensor and replaces

$$\widetilde{A}_t \;=\; \widehat{A}_t + \mathbf{1}\{k \ge k_{\mathrm{on}}\} f_\phi(s_t, a_t).$$

The attacker cannot edit the transition, sampled action, policy parameters, critic target, optimizer state, deployment policy, or downstream telemetry. The perturbation is detached before the actor loss is differentiated; its effect must therefore travel through the estimated advantage and the normal PPO update. The critic target and raw TD/GAE stream remain clean.

The distinction from an unrestricted attacker matters. With full access to the training stack, an adversary can directly write the desired weights or telemetry, so the interesting question is not whether the attack is undetectable. Here the narrower question is whether the loss of performance is difficult to distinguish from ordinary training noise until training has ended, even though the final frozen policy is suboptimal or exhibits a trigger-specific action change. Experiment 7 operationalizes this question with an estimator-only manipulation while keeping rewards, observations, sampled actions, and optimizer state unchanged.

For a target state–action pair, f_ϕ creates the action dependence needed to change a policy update even though the environment trajectory is unchanged. The mechanism studied here is therefore

$$\boxed{\text{advantage-level perturbation} \longrightarrow \text{distorted actor estimator} \longrightarrow \text{gradient perturbation} \longrightarrow \text{policy behavior.}}$$

Prior work has established the feasibility of test-time policy attacks and training-time reward, transition, and data poisoning (Huang et al., 2017; Gleave et al., 2020; Ma et al., 2019; Zhang et al., 2020; Rakhsha et al., 2021). The present study is narrower and diagnostic: it identifies which part of an advantage-estimator perturbation reaches the actor update, proves a stationarity bound for that shift, and tests whether the resulting training behavior is reflected in final policy behavior. The result is not a new PPO algorithm, not a claim that estimator bias improves performance, and not a complete convergence theorem for PPO-Clip. A TD-level realization is given only in Appendix A.

2 Setup and Notation

We consider a discounted Markov decision process with discount factor $\gamma \in (0,1)$ and a differentiable stochastic policy

$$\pi_\theta(a \mid s), \qquad \theta \in \mathbb{R}^d.$$

The clean policy objective is

$$J(\theta) := \mathbb{E}_{\pi_\theta}\left[\sum_{t=0}^{\infty} \gamma^t r_t\right].$$

For a policy π_θ, define

$$Q^{\pi_\theta}(s,a) := \mathbb{E}_{\pi_\theta}\left[\sum_{\ell=0}^{\infty} \gamma^\ell r_{t+\ell} \,\middle|\, s_t = s,\ a_t = a\right],$$

$$V^{\pi_\theta}(s) := \mathbb{E}_{a\sim\pi_\theta(\cdot|s)}\left[Q^{\pi_\theta}(s,a)\right], \qquad A^{\pi_\theta}(s,a) := Q^{\pi_\theta}(s,a) - V^{\pi_\theta}(s).$$

The policy score is

$$\boxed{z_\theta(s,a) := \nabla_\theta \log \pi_\theta(a \mid s).}$$

Under the policy-gradient theorem,

$$\boxed{\nabla J(\theta) = \mathbb{E}_{s,a}\left[A^{\pi_\theta}(s,a) z_\theta(s,a)\right],}$$

where actions are sampled from π_θ and the state expectation uses the corresponding discounted on-policy visitation distribution.

The TD error formed with a value function V is

$$\delta_t := r_t + \gamma(1-d_t)V(s_{t+1}) - V(s_t).$$

Write $q_\lambda \coloneqq \gamma\lambda < 1$ for the GAE discount and define

$$\widehat{A}_t \coloneqq \sum_{\ell=0}^{\infty} q_\lambda^\ell \delta_{t+\ell}.$$

For finite episodes, the sum terminates at the episode boundary and the terminal bootstrap is zero.

Assumption 1 (Correctness of the unclipped GAE)**.** *The unclipped estimator satisfies*

$$\mathbb{E}[\widehat{A}_t \mid s_t = s, a_t = a] = A^{\pi_\theta}(s, a).$$

This holds, for example, when $V = V^{\pi_\theta}$, terminal handling is correct, and the post-action trajectory follows π_θ. The later convergence result only needs the displayed conditional-mean identity.

For example, with $V = V^{\pi_\theta}$, one has $\mathbb{E}[\delta_t \mid s_t = s, a_t = a] = A^{\pi_\theta}(s, a)$, while the conditional mean of each later on-policy TD error is zero after averaging over the action drawn from π_θ. This yields the displayed GAE identity provided the relevant sums are integrable and terminal bootstrapping is handled correctly.

When V is learned rather than exact, critic approximation and truncation effects must either be controlled separately or included in the generic perturbation f_θ below; the theorem does not silently attribute those effects to clipping.

3 Advantage Estimator Perturbations and Action-Dependent Bias

3.1 Asymmetric clipping as a special case

For thresholds $c_+ > 0$ and $c_- > 0$, define the positive and negative tail excesses

$$b_{+,t} \coloneqq (\delta_t - c_+)_+, \qquad b_{-,t} \coloneqq (-\delta_t - c_-)_+.$$

The two one-sided clipping rules are

$$\boxed{\delta_t^+ = \min\{\delta_t, c_+\} = \delta_t - b_{+,t},}$$

and

$$\boxed{\delta_t^- = \max\{\delta_t, -c_-\} = \delta_t + b_{-,t}.}$$

If both tails are clipped, the resulting TD error is

$$\boxed{\delta_t^{\text{both}} = \min\{c_+, \max\{-c_-, \delta_t\}\} = \delta_t - b_{+,t} + b_{-,t}.}$$

The corresponding GAE estimators satisfy

$$\widehat{A}_t^+ = \widehat{A}_t - \sum_{\ell=0}^{\infty} q_\lambda^\ell b_{+,t+\ell},$$

$$\widehat{A}_t^- = \widehat{A}_t + \sum_{\ell=0}^{\infty} q_\lambda^\ell b_{-,t+\ell},$$

and

$$\widehat{A}_t^{\text{both}} = \widehat{A}_t - \sum_{\ell=0}^{\infty} q_\lambda^\ell b_{+,t+\ell} + \sum_{\ell=0}^{\infty} q_\lambda^\ell b_{-,t+\ell}.$$

Define the conditional clipping biases

$$\boxed{B_+(s,a) := \sum_{\ell=0}^{\infty} q_\lambda^\ell \mathbb{E}[b_{+,t+\ell} \mid s_t = s, a_t = a],}$$

and

$$\boxed{B_-(s,a) := \sum_{\ell=0}^{\infty} q_\lambda^\ell \mathbb{E}[b_{-,t+\ell} \mid s_t = s, a_t = a].}$$

By Assumption 1,

$$\boxed{\mathbb{E}[\widehat{A}_t^+ \mid s_t = s, a_t = a] = A^{\pi_\theta}(s,a) - B_+(s,a),}$$

$$\boxed{\mathbb{E}[\widehat{A}_t^- \mid s_t = s, a_t = a] = A^{\pi_\theta}(s,a) + B_-(s,a).}$$

Thus upper clipping is pessimistic and lower clipping is optimistic at the conditional advantage-estimate level. For joint clipping,

$$\mathbb{E}[\widehat{A}_t^{\text{both}} \mid s_t = s, a_t = a] = A^{\pi_\theta}(s,a) - B_+(s,a) + B_-(s,a).$$

3.2 Tail representation

For an integrable random variable X and threshold c,

$$\mathbb{E}[(X-c)_+] = \int_c^\infty \mathbb{P}(X > u)\, du.$$

Tonelli's theorem therefore gives

$$\boxed{B_+(s,a) = \sum_{\ell=0}^{\infty} q_\lambda^\ell \int_{c_+}^{\infty} \mathbb{P}(\delta_{t+\ell} > u \mid s_t = s, a_t = a)\, du,}$$

and

$$\boxed{B_-(s,a) = \sum_{\ell=0}^{\infty} q_\lambda^\ell \int_{c_-}^{\infty} \mathbb{P}(\delta_{t+\ell} < -u \mid s_t = s, a_t = a)\, du.}$$

Increasing a threshold suppresses the corresponding tail contribution for a fixed TD-error distribution. In an evolving learner, the distribution also changes with the policy; the empirical threshold sweep measures that additional effect rather than treating it as a theorem consequence.

3.3 Only action-dependent bias reaches the expected actor update

The main experiments use an advantage-level perturbation. Let $\widehat{A}_t$ denote the clean advantage estimator produced by GAE and let $f_\phi(s_t, a_t)$ be the detached payload added immediately before the actor update:

$$\widehat{A}_t^\phi = \widehat{A}_t + f_\phi(s_t, a_t).$$

The theoretical identity below is written in raw advantage units. If PPO uses fixed-reference normalization $\mathcal{N}_{\text{ref}}(x) = (x - \mu_A)/\sigma_A$, the same transformation is applied to both streams, so the actor-facing perturbation is f_ϕ/σ_A. Self-normalization of the poisoned stream is not used, because it would remove part of the shared shift being measured. The critic target is unchanged.

Suppose the conditional mean of the poisoned estimator is

$$\mathbb{E}[\widehat{A}_t^{\phi} \mid s_t = s, a_t = a] = A^{\pi_\theta}(s,a) + f_\theta(s,a),$$

where $f_\theta(s,a) = \mathbb{E}[f_\phi(s_t,a_t) \mid s_t = s, a_t = a]$. Define

$$g_\phi(\theta) \coloneqq \mathbb{E}[\widehat{A}_t^{\phi} z_\theta(s_t,a_t)], \qquad e_\phi(\theta) \coloneqq g_\phi(\theta) - \nabla J(\theta).$$

Then

$$e_\phi(\theta) = \mathbb{E}[f_\theta(s,a) z_\theta(s,a)].$$

Let

$$\bar{f}_\theta(s) \coloneqq \mathbb{E}_{a\sim\pi_\theta(\cdot|s)}[f_\theta(s,a)], \qquad \Delta f_\theta(s,a) \coloneqq f_\theta(s,a) - \bar{f}_\theta(s).$$

Since

$$\mathbb{E}_{a\sim\pi_\theta(\cdot|s)}\left[z_\theta(s,a)\right] = \mathbb{E}_{a\sim\pi_\theta(\cdot|s)}\left[\nabla_\theta \log \pi_\theta(a \mid s)\right] = 0,$$

we obtain

$$\boxed{e_\phi(\theta) = \mathbb{E}[\Delta f_\theta(s,a) z_\theta(s,a)].}$$

The identity concerns the expectation of the estimator, not a derivative of the detached payload. Any zero-mean residual $f_\phi - f_\theta$ is part of the stochastic-noise term in the empirical update.

For upper clipping, $f_+ = -B_+$, so define

$$C_+^\theta(s) = \mathbb{E}_{a\sim\pi_\theta}[B_+(s,a)], \qquad \Delta B_+(s,a) = B_+(s,a) - C_+^\theta(s),$$

and obtain

$$\boxed{e_+(\theta) = -\mathbb{E}[\Delta B_+(s,a) z_\theta(s,a)].}$$

For lower clipping, define C_-^θ and ΔB_- analogously. Then

$$\boxed{e_-(\theta) = \mathbb{E}[\Delta B_-(s,a) z_\theta(s,a)].}$$

For joint clipping,

$$e_{\text{both}}(\theta) = \mathbb{E}[(-\Delta B_+(s,a) + \Delta B_-(s,a)) z_\theta(s,a)].$$

Assume the score is uniformly bounded,

$$\|z_\theta(s,a)\| \le G_z,$$

and define

$$\varepsilon_B^+ \coloneqq \sup_{\theta,s,a} \left|\Delta B_+(s,a)\right|, \qquad \varepsilon_B^- \coloneqq \sup_{\theta,s,a} \left|\Delta B_-(s,a)\right|.$$

Then

$$\boxed{\|e_+(\theta)\| \le G_z \varepsilon_B^+, \qquad \|e_-(\theta)\| \le G_z \varepsilon_B^-.}$$

For joint clipping, the rigorous triangle-inequality bound is

$$\boxed{\|e_{\text{both}}(\theta)\| \le G_z(\varepsilon_B^+ + \varepsilon_B^-).}$$

The relevant quantity is therefore within-state variation, not the total positive or negative clipping bias.

3.4 Explicit tail bounds

Suppose uniformly over θ, states, actions, and ℓ that

$$\mathbb{P}(\delta_{t+\ell} > u \mid s, a) \le \psi_+(u), \qquad \mathbb{P}(\delta_{t+\ell} < -u \mid s, a) \le \psi_-(u),$$

where the tail majorants are integrable on the relevant domains. Then

$$R_+(c_+) \coloneqq \frac{1}{1-\gamma\lambda} \int_{c_+}^{\infty} \psi_+(u)\, du, \qquad R_-(c_-) \coloneqq \frac{1}{1-\gamma\lambda} \int_{c_-}^{\infty} \psi_-(u)\, du,$$

give

$$0 \le B_+(s,a) \le R_+(c_+), \qquad 0 \le B_-(s,a) \le R_-(c_-).$$

Because each $C_\pm^\theta(s)$ is an average of a quantity in the same interval,

$$\boxed{\varepsilon_B^+ \le R_+(c_+), \qquad \varepsilon_B^- \le R_-(c_-).}$$

Consequently,

$$\|e_+(\theta)\| \le G_z R_+(c_+), \qquad \|e_-(\theta)\| \le G_z R_-(c_-).$$

If instead $|\delta_t| \le D_\delta$ almost surely, then

$$\boxed{\varepsilon_B^+ \le \frac{(D_\delta - c_+)_+}{1-\gamma\lambda}, \qquad \varepsilon_B^- \le \frac{(D_\delta - c_-)_+}{1-\gamma\lambda}.}$$

Thus a threshold beyond the corresponding TD-error support makes that clipping operation inactive and its gradient perturbation zero.

4 Convergence of the Biased Policy-Gradient Update

The convergence statement is written for a generic estimator perturbation; the upper, lower, and joint clipping cases follow by substituting the bounds above. This avoids proving the same inequality three times.

Theorem 1 (Finite-time stationarity under bounded estimator bias)**.** *Let J be L-smooth for some $L > 0$ and bounded above by $J_{\max}$, and let θ_0 be deterministic. Consider the ascent iteration*

$$\theta_{k+1} = \theta_k + \alpha_k \widehat{g}_k, \qquad \widehat{g}_k = \nabla J(\theta_k) + e_k + \xi_k,$$

where $e_k = e(\theta_k)$ is θ_k-measurable and, for every k,

$$\|e_k\| \le \varepsilon_g, \qquad \mathbb{E}[\xi_k \mid \theta_k] = 0, \qquad \mathbb{E}[\|\xi_k\|^2 \mid \theta_k] \le \sigma^2.$$

If $0 < \alpha_k \le 1/(4L)$, then for every $T \ge 1$,

$$\boxed{\frac{\sum_{k=0}^{T-1} \alpha_k \mathbb{E}[\|\nabla J(\theta_k)\|^2]}{\sum_{k=0}^{T-1} \alpha_k} \le \frac{2[J_{\max} - J(\theta_0)]}{\sum_{k=0}^{T-1} \alpha_k} + \frac{5}{2}\varepsilon_g^2 + L\sigma^2 \frac{\sum_{k=0}^{T-1} \alpha_k^2}{\sum_{k=0}^{T-1} \alpha_k}.}$$

Proof. Write $\mathbb{E}_k[\cdot] \coloneqq \mathbb{E}[\cdot \mid \theta_k]$ and $g_k = \nabla J(\theta_k)$. L-smoothness gives the ascent lower bound

$$\begin{aligned}\mathbb{E}_k[J(\theta_{k+1})] \geq J(\theta_k) + \alpha_k \|g_k\|^2 + \alpha_k \langle g_k, e_k\rangle \\ - \frac{L}{2}\alpha_k^2 \mathbb{E}_k[\|g_k + e_k + \xi_k\|^2].\end{aligned}$$

The conditional mean-zero property implies

$$\mathbb{E}_k[\|g_k + e_k + \xi_k\|^2] \leq \|g_k + e_k\|^2 + \sigma^2.$$

Using

$$\langle g_k, e_k\rangle \geq -\frac{1}{4}\|g_k\|^2 - \|e_k\|^2, \qquad \|g_k + e_k\|^2 \leq 2\|g_k\|^2 + 2\|e_k\|^2,$$

we obtain

$$\begin{aligned}\mathbb{E}_k[J(\theta_{k+1})] \geq J(\theta_k) + \left(\frac{3}{4}\alpha_k - L\alpha_k^2\right)\|g_k\|^2 \\ - \left(\alpha_k + L\alpha_k^2\right)\|e_k\|^2 - \frac{L}{2}\alpha_k^2\sigma^2.\end{aligned}$$

Since $\alpha_k \leq 1/(4L)$,

$$\frac{3}{4}\alpha_k - L\alpha_k^2 \geq \frac{1}{2}\alpha_k, \qquad \alpha_k + L\alpha_k^2 \leq \frac{5}{4}\alpha_k.$$

Therefore,

$$\mathbb{E}_k[J(\theta_{k+1})] \geq J(\theta_k) + \frac{\alpha_k}{2}\|g_k\|^2 - \frac{5\alpha_k}{4}\varepsilon_g^2 - \frac{L}{2}\alpha_k^2\sigma^2.$$

Taking total expectations, summing from 0 to $T-1$, and using $\mathbb{E}[J(\theta_T)] \leq J_{\max}$ gives

$$\frac{1}{2}\sum_{k=0}^{T-1}\alpha_k \mathbb{E}[\|g_k\|^2] \leq J_{\max} - J(\theta_0) + \frac{5}{4}\varepsilon_g^2 \sum_{k=0}^{T-1}\alpha_k + \frac{L}{2}\sigma^2 \sum_{k=0}^{T-1}\alpha_k^2.$$

Multiplication by 2 and division by $\sum_{k=0}^{T-1}\alpha_k$ yields the claimed bound. □

Proof audit. The proof uses only the displayed smoothness inequality, the conditional mean-zero and second-moment conditions on ξ_k, the uniform bias bound, and the upper bound on J. It establishes a weighted-average bound on $\mathbb{E}[\|\nabla J(\theta_k)\|^2]$; it does not establish convergence of every parameter iterate.

Corollary 1 (Upper, lower, and joint clipping)**.** *Under the conditions of Theorem 1, the clipping cases satisfy the theorem with*

$$\varepsilon_g^+ = G_z \varepsilon_B^+, \qquad \varepsilon_g^- = G_z \varepsilon_B^-,$$

and, for joint clipping,

$$\varepsilon_g^{\text{both}} = G_z(\varepsilon_B^+ + \varepsilon_B^-).$$

For example, upper clipping gives

$$\boxed{\frac{\sum_{k=0}^{T-1}\alpha_k \mathbb{E}[\|\nabla J(\theta_k)\|^2]}{\sum_{k=0}^{T-1}\alpha_k} \leq \frac{2[J_{\max} - J(\theta_0)]}{\sum_{k=0}^{T-1}\alpha_k} + \frac{5}{2}G_z^2(\varepsilon_B^+)^2 + L\sigma^2 \frac{\sum_{k=0}^{T-1}\alpha_k^2}{\sum_{k=0}^{T-1}\alpha_k}.}$$

The lower-clipping formula replaces ε_B^+ by ε_B^-. For joint clipping, the clipping term is

$$\frac{5}{2}G_z^2(\varepsilon_B^+ + \varepsilon_B^-)^2.$$

For a constant step size α,

$$\boxed{\frac{1}{T}\sum_{k=0}^{T-1}\mathbb{E}[\|\nabla J(\theta_k)\|^2] \le \frac{2[J_{\max} - J(\theta_0)]}{\alpha T} + L\alpha\sigma^2 + \frac{5}{2}\varepsilon_g^2.}$$

For diminishing step sizes satisfying

$$\sum_{k=0}^{\infty}\alpha_k = \infty, \qquad \sum_{k=0}^{\infty}\alpha_k^2 < \infty, \qquad \alpha_k \le \frac{1}{4L},$$

$$\boxed{\limsup_{T\to\infty}\frac{\sum_{k=0}^{T-1}\alpha_k\mathbb{E}[\|\nabla J(\theta_k)\|^2]}{\sum_{k=0}^{T-1}\alpha_k} \le \frac{5}{2}\varepsilon_g^2.}$$

Thus the theorem gives exact weighted-average stationarity when $\varepsilon_g = 0$, and otherwise gives an $O(\varepsilon_g)$ stationary neighborhood in gradient norm. It does not imply that a small gradient of the biased update corresponds to a high-return policy.

5 Relation to PPO and the Security Interpretation

The theorem analyzes an ascent update driven by an advantage-based policy gradient. PPO additionally uses the probability ratio

$$\rho_\theta(s,a) := \frac{\pi_\theta(a \mid s)}{\pi_{\theta_{\text{old}}}(a \mid s)}$$

and the clipped surrogate

$$L^{\text{PPO}}(\theta) = \mathbb{E}\left[\min\left\{\rho_\theta A, \operatorname{clip}(\rho_\theta, 1-\epsilon, 1+\epsilon)A\right\}\right].$$

Ratio clipping introduces a second, optimizer-specific nonlinearity. The theorem above is therefore not a complete convergence theorem for PPO-Clip; it isolates the additional perturbation caused by the advantage estimator. In the experiments, the vanilla policy-gradient quantity is logged as a theorem-aligned diagnostic, separately from the PPO surrogate gradient used to update the policy.

The security interpretation is also deliberately limited. A small biased update can appear stationary because

$$g_f(\theta) = \nabla J(\theta) + e_f(\theta) \approx 0$$

even when $\|\nabla J(\theta)\|$ is nonzero. This is an estimator-level stationarity claim, not a claim that the final policy is optimal or that an attacker is invisible to every audit. Conventional PPO logs may fail to separate clean and attacked runs during training, while an estimator-aware audit can expose the transformed advantage values directly. This empirical stealth claim is tested separately from the theorem. Experiments 5–7 test stationarity, telemetry, targeted action change, and persistence after the perturbation is removed.

6 Experimental Validation

6.1 Scope and common estimator

The experiments study the generalized estimator perturbation at the advantage level, not clipping as a standalone algorithm. Clipping remains in the theory as a special case and may be reported as an optional reference row, but it is not the primary experimental treatment. The clean TD/GAE computation is

$$\delta_t = r_t + \gamma(1 - d_t)V_\omega(s_{t+1}) - V_\omega(s_t), \qquad \widehat{A}_t = \mathrm{GAE}(\delta_{t:T}).$$
$$\widetilde{A}_t = \widehat{A}_t + \mathbf{1}\{k \geq k_{\mathrm{on}}\} f_\phi(s_t, a_t).$$

Here f_ϕ is the raw advantage-level payload in the same units as $\widehat{A}_t$. It is detached from differentiation. The critic target and raw TD/GAE stream remain clean. Thus the only attacker-controlled quantity is the advantage tensor supplied to the actor update; the environment data, sampled actions, policy parameters, optimizer, and deployment policy are not directly modified. If fixed-reference normalization is used, it is applied identically to both streams, so the effective actor perturbation is f_ϕ/σ_A. The theorem-aligned gradient diagnostics use the raw advantage streams; PPO training may use this fixed common normalization.

Let $\widetilde{o}_t$ be a normalized observable state and let

$$h(\widetilde{o}) = \sqrt{2}\cos(w^\top \widetilde{o} + b), \qquad g_\pi(s, a) = q(a) - \mathbb{E}_{a' \sim \pi(\cdot|s)}[q(a')]$$

be fixed state and action features. The main mechanism family is

$$\boxed{f_{\rho,\beta}(s, a) = \rho\, \sigma_A\, h(\widetilde{o})\, [(1 - \beta) + \beta g_\pi(s, a)]\,,}$$

where ρ is the calibrated dose, $\beta \in \{0, .25, .5, 1\}$ controls action selectivity, and σ_A is frozen from clean calibration data. The action feature is centered exactly under the current policy:

$$\mathbb{E}_{a \sim \pi(\cdot|s)}[g_\pi(s, a)] = 0.$$

For the binary targeted attack, use a fixed-direction payload

$$\boxed{f_{\mathcal{T},a^\dagger}(s, a) = \rho\, \sigma_A\, \mathbf{1}\{o(s) \in \mathcal{T}\} \left[\mathbf{1}\{a = a^\dagger\} - \pi(a^\dagger \mid s)\right].}$$

This payload increases the estimated advantage weight of $a^\dagger$ and decreases the competing action on every trigger state. The trigger set $\mathcal{T}$ uses only observable inputs available to the advantage-producing component; it cannot use the hidden NutriRL target or digestion backlog.

6.2 Environments and common calibration

CartPole-v1 is the fast binary-action mechanism check. NutriRL is the primary domain case study: it presents a binary consume/skip decision, with delayed heterogeneous nutrient absorption, hidden targets, and irreversible accumulation. These properties make the two attack directions interpretable: $a^\dagger = 1$ induces over-consumption and overshoot, whereas $a^\dagger = 0$ induces under-consumption and target shortfall (Khan et al., 2026).

For each environment, train clean PPO reference policies with five seeds, freeze the actor, critic, and normalization statistics, and collect a clean calibration set. Freeze σ_A, the normalized feature map, h, the trigger definition, and all selected doses before poisoned training. A trigger is defined from observable state and food identity only. Fresh episodes are used for every final evaluation.

6.3 Experiment 1: Reference policy and calibration

Objective. Establish the clean learning reference, observation normalization, TD/GAE scale, and fixed evaluation splits. This is a calibration experiment, not a test of the bias hypothesis.

Execution. Train clean PPO for five seeds per environment. Freeze the final policies and collect the calibration trajectories used by Experiments 2–4.

Log and plot. Record clean return, episode length, TD and advantage scales, and action frequencies. Plot the clean learning curves and the reference TD/advantage distributions. Report training time only as the fixed budget used by all subsequent runs; final policy evaluation is inference-only.

Expected result in our favor. The clean policies should learn reliably and reach a stable performance range across seeds, with a fixed and reproducible TD/advantage scale.

What this establishes. This experiment does not test the perturbation hypothesis. It establishes the empirical clean reference required by Assumption 1, provides the clean score-gradient baseline, and fixes the scale against which later perturbation magnitudes are measured. If the clean reference is unstable, later bias or attack results are not interpretable.

6.4 Experiment 2: Perturbation-library sanity and centering

Objective. Verify the construction of $f_{\rho,\beta}$ before any policy claim:

$$\mathbb{E}_\pi[g_\pi(s,a)] = 0, \qquad \frac{\operatorname{std}(f_{\rho,\beta})}{\operatorname{std}(\widehat{A})} \approx \rho$$

under the frozen calibration distribution. The experiment also verifies that $\beta = 0$ has no direct same-step action-selective component.

Execution. Use frozen reference trajectories and evaluate all actions exactly for discrete policies. Compute action-conditional means of f, its standard deviation, $\mathbb{E}|f|$, and the within-state action-dependent component

$$\Delta f(s,a) = f(s,a) - \mathbb{E}_{a'\sim\pi}[f(s,a')].$$

No policy training is performed.

Log and plot. Plot dose linearity, action-conditional perturbation means, and $\operatorname{std}(\Delta f)$ versus β. Failure of centering or dose calibration stops the later experiments.

Expected result in our favor. The measured action mean of g_π should be numerically zero, $\operatorname{std}(f)/\operatorname{std}(\widehat{A})$ should be approximately ρ, and $\operatorname{std}(\Delta f)$ should be near zero at $\beta = 0$ and increase with β.

What this establishes. This validates that the function family actually separates total perturbation from within-state action-dependent perturbation. It directly operationalizes

$$e_f(\theta) = \mathbb{E}[\Delta f_\theta(s,a) z_\theta(s,a)]$$

from the action-dependent-bias theory. A positive result means later gradient effects can be attributed to the designed f, not accidental scaling or imperfect centering. It is not evidence of performance improvement.

6.5 Experiment 3: Dose pilot

Objective. Select three non-saturating perturbation doses before the main grid. The selection is based on realized estimator magnitude and action probability movement, not final return.

Execution. Use $\beta = 1$ and a pre-registered ladder such as

$$\rho \in \{0.05, 0.10, 0.25, 0.50, 1.00\}.$$

Run two seeds per dose on CartPole and NutriRL. Select three positive doses that produce measurable but non-degenerate perturbations; retain $\rho = 0$ as the clean condition. The selection rule is frozen before Experiment 4.

Log and plot. Log $\mathrm{std}(f)$, $\mathrm{std}(\Delta f)$, gradient perturbation norm, cosine with the clean shadow gradient, and action probability movement. Plot these quantities against ρ. This is a training-time pilot; the final action and return measurements are inference-time measurements and are not used for dose selection.

Expected result in our favor. The perturbation norm and action-probability movement should increase smoothly with ρ, while the selected doses should avoid policy collapse, probability saturation, or nearly random behavior.

What this establishes. This identifies the measurable regime in which the theoretical bias parameter ε_g is nonzero but the PPO experiment remains a learning experiment rather than a numerical failure. It provides the practical mapping

$$\rho \longmapsto \mathrm{std}(\Delta f) \longmapsto \|\widehat{e}_\phi\|\,.$$

If no dose produces a measurable perturbation, the attack claim is untestable at the chosen scale; if every dose collapses the policy, the mechanism is too strong to support a stealth or persistence claim.

6.6 Experiment 4: Mechanism grid

Objective. Test whether increasing action selectivity and dose produces increasing within-state estimator bias and gradient distortion, while separating this effect from the state-only component.

Execution. For each environment, run

$$3\ \rho\text{-values} \times 4\ \beta\text{-values} \times 5 \text{ seeds} = 60$$

perturbed runs, plus five clean runs. The total is therefore **65 runs per environment**. The four β values are $\{0, .25, .5, 1\}$; the three positive ρ values are selected in Experiment 3. The clean GAE computation is completed first, and $f_{\rho,\beta}$ is then added to the advantage tensor immediately before the actor update. The critic target is clean.

Primary measurements. At fixed checkpoints compute

$$\widehat{e}_\phi = \frac{1}{N}\sum_i \left(\widetilde{A}_i - \widehat{A}_i\right) z_i, \qquad z_i = \nabla_\theta \log \pi_\theta(a_i \mid s_i),$$

along with $\|\widehat{e}_\phi\|$, cosine with the clean shadow update, $\mathrm{std}(f)$, $\mathrm{std}(\Delta f)$, and update variance.

Log and plot. Plot gradient perturbation versus β and ρ, action-dependent versus total perturbation, and update variance. Because the perturbation is added after GAE, the $\beta = 0$ state-only cancellation is tested directly without an additional temporal-convolution effect.

Expected result in our favor. At fixed ρ, $\|\widehat{e}_\phi\|$ should generally increase with β, with the smallest direct action-conditioned effect at $\beta = 0$ and the largest at $\beta = 1$. Increasing ρ should increase the effect monotonically over the selected dose range. The gradient cosine should decrease in the same regime, while update variance should be reported separately rather than treated as bias.

What this establishes. This is the main empirical test of the theory's decomposition:

$$\text{total estimator perturbation} \;=\; \text{state/shared component} \;+\; \Delta f(s,a), \qquad e_f = \mathbb{E}[\Delta f\, z].$$

A positive result shows that action-selective variation, rather than merely large estimator magnitude, reaches the policy-gradient direction. Because f is added after GAE, the state-only cancellation argument applies directly to the same-step score expectation. The observed quantity is an empirical gradient perturbation, not equality with the theorem's worst-case bound.

6.7 Experiment 5: Targeted online advantage manipulation

Objective. Test whether a compromised advantage-producing component can produce a targeted change in the learned policy without modifying the environment, action sampler, policy, optimizer, or critic target.

Execution. Use NutriRL as the primary environment and CartPole as the binary control. For NutriRL, define an observable trigger using a held-out food identity and an observed nutrient-state interval. Run two target directions:

$$a^\dagger = 1 \quad (\text{consume}), \qquad a^\dagger = 0 \quad (\text{skip}).$$

Compare clean PPO, targeted $f_{\mathcal{T},a^\dagger}$, and a random-trigger control with the same dose and activation frequency. A reverse-target condition may be retained only as a signed control, not as a second attack definition. Use five matched seeds per arm and direction. Disable the perturbation during final evaluation.

Primary inference metrics. For a held-out trigger state, define the binary action logit

$$m_\theta(s) = \log \frac{\pi_\theta(1 \mid s)}{\pi_\theta(0 \mid s)}.$$

The targeted flip rate is

$$\mathrm{TFR} = \mathbb{P}_{s\in\mathcal{T}}\left[a_{\text{clean}}(s) \neq a^\dagger \;\wedge\; a_{\text{poison}}(s) = a^\dagger\right].$$

For NutriRL, also report overshoot probability and nutrient target shortfall for the consume and skip directions, respectively. Report collateral flips on held-out non-trigger states.

Log and plot. Plot trigger versus non-trigger action probabilities, targeted flip rate, collateral flip rate, margin change, overshoot/shortfall, and final return. This is an online training experiment followed by frozen-policy inference; the environmental consequences are evaluated only after training.

Expected result in our favor. For each target direction, the targeted perturbation should produce a higher held-out trigger-state target-action probability and targeted flip rate than both clean PPO and the random-trigger control, while collateral flips remain substantially smaller. In NutriRL, the consume-target should increase overshoot and the skip-target should increase target shortfall.

What this establishes. This is the causal, policy-level demonstration:

$$f_{\mathcal{T},a^\dagger} \longrightarrow \widetilde{\delta} \longrightarrow \widetilde{A} \longrightarrow \text{targeted action change.}$$

It turns the abstract gradient-bias term e_f into an observable policy manipulation under the restricted threat model. The random-trigger control shows that the effect depends on the chosen observable state pattern rather than merely on injecting extra estimator noise. It does not show that bias is beneficial to the learner; it shows why the bias can be useful to an attacker.

6.8 Experiment 6: Stationarity and telemetry audit

Objective. Test separately whether the poisoned update can appear stationary and whether conventional PPO telemetry distinguishes it during training. The stationarity theorem does not imply telemetry invisibility, so this is an empirical test.

Execution. Reuse the runs from Experiments 4–5. At checkpoints, compute a clean shadow update from raw advantages on the same batch and compare it with the update produced by $\widetilde{A}$. Use the vanilla score-gradient diagnostic separately from the PPO surrogate gradient.

Measurements. Report poisoned and clean-shadow gradient norms, their cosine, the norm of their difference, policy target-action probability, and running late-training averages. For the standard audit, train a detector using only conventional logs:

`policy_loss, value_loss, entropy, approx_kl, clip_fraction, gradient_norm, episode_return`.

Split by complete training run, not by individual updates, and report standard-log AUROC at several checkpoints. An estimator-aware audit may use raw/transformed TD error and f_ϕ diagnostics; it is expected to expose the manipulation.

Log and plot. Plot clean versus poisoned stationarity curves, gradient cosine, and standard-log AUROC over training. Phrase the result as weak or delayed discriminability against the specified telemetry, never as universal stealth.

Expected result in our favor. During late poisoned training, the poisoned score-gradient estimate should be small or appear stationary, while the clean-shadow estimate from the same batch remains measurably different. Standard-log AUROC should remain near chance or become discriminative only later, whereas the estimator-aware audit should separate the runs immediately.

What this establishes. The gradient comparison tests the stationarity mechanism

$$g_f(\theta) = \nabla J(\theta) + e_f(\theta) \approx 0 \quad \text{while} \quad \|\nabla J(\theta)\| > 0.$$

The standard-log detector tests the separate empirical stealth claim. A positive result supports the statement that stationarity of the poisoned update is not equivalent to clean optimality and that ordinary PPO telemetry may not reveal the cause early. It does not prove universal log indistinguishability; if standard logs separate the runs, the correct conclusion is that this attack is detectable under that telemetry.

6.9 Experiment 7: Persistence after the perturbation is removed

Objective. Determine whether an online advantage perturbation produces a persistent inference-time policy change rather than only a transient training effect.

Execution. At the end of the attack window, save poisoned checkpoints and matched clean checkpoints at the same training step. Turn f_ϕ off and continue both conditions for the same number of updates with identical seeds and budgets. Evaluate at the attack endpoint and throughout the clean-continuation phase.

Measurements. Track targeted flip rate, target-action probability, logit-margin change, collateral flip rate, return, and NutriRL overshoot/shortfall. Report the decay of the poisoned–clean difference as a function of clean continuation updates. The persistence result is positive only if the poisoned checkpoint differs from the matched clean continuation beyond ordinary training drift.

Log and plot. Plot flip-rate and environment-consequence trajectories after attack removal, with matched clean continuation as the baseline. This is the final inference-time validation of the training-time estimator manipulation.

Expected result in our favor. After f_ϕ is disabled, the poisoned policy should retain a measurable targeted flip-rate and corresponding NutriRL over/under-regulation for at least part of the clean-continuation window, while the matched clean policy should not show the same change. The poisoned–clean gap may decay, but it should not vanish immediately.

What this establishes. This distinguishes a persistent training-time policy manipulation from a temporary change in one PPO update. It connects the stationarity result to deployment behavior: the estimator bias changes the learned parameters through ordinary optimization, and the resulting policy remains altered even when the estimator path is restored. Immediate disappearance is a valid negative result: it means the perturbation affects transient updates but does not create a persistent policy change.

6.10 Final required experiment list

The required causal chain is

$$\boxed{f(s,a) \longrightarrow \widetilde{A} \longrightarrow \text{biased update} \longrightarrow \text{targeted final behavior.}}$$

No experiment should claim that the perturbation improves PPO or constitutes a safety objective. The paper's result is that a restricted advantage-estimator compromise can create measurable action-dependent learning distortion and, in a delayed irreversible environment, directional under- or over-regulation.

Table 1: Final experiments required for the paper.

ID	Goal	Mode	Expected result and theory established
E1	Clean reference and calibration	Training, then frozen inference	Stable clean baseline; fixes the empirical reference for Assumption 1.
E2	Verify f centering and dose construction	Frozen/offline	$\mathbb{E}_\pi[g] = 0$ and Δf increases with β; validates the bias decomposition.
E3	Select non-saturating doses	Short online training	Measurable $\rho \mapsto \|\widehat{e}_\phi\|$ without policy collapse; calibrates the theorem's bias regime.
E4	Establish mechanism across β and ρ	Matched online PPO	$\|\widehat{e}_\phi\|$ grows with action selectivity; tests $e_f = \mathbb{E}[\Delta f z]$.
E5	Demonstrate targeted advantage manipulation	Online attack, then inference	Targeted flips and NutriRL over/under-regulation exceed controls; validates the causal attack path.
E6	Test stationarity and conventional-log audit	Training diagnostics	Poisoned update appears stationary while shadow update differs; tests the stationarity mechanism and telemetry claim separately.
E7	Test persistence after attack removal	Matched continuation, then inference	Flip and environmental effects persist beyond immediate training; establishes lasting policy alteration.

A TD-level realization of the advantage poison

The main experiments inject the payload after GAE, at the advantage tensor consumed by the actor. This appendix formulates the corresponding TD-level version. It is an alternative realization, not an additional primary experimental condition.

Let $p_\phi(t)$ denote a payload added to the raw TD residual:

$$\widetilde{\delta}_t = \delta_t + p_\phi(t), \qquad \widetilde{A}_t = \mathrm{GAE}(\widetilde{\delta}_{t:T}).$$

Let $m_{t+1} \in \{0, 1\}$ indicate that the transition from t to $t+1$ does not cross an episode boundary, and write $q_\lambda = \gamma\lambda$. The difference between the poisoned and clean GAE streams is

$$u_\phi(t) := \widetilde{A}_t - \widehat{A}_t = p_\phi(t) + q_\lambda m_{t+1} u_\phi(t+1),$$

or, equivalently,

$$\boxed{u_\phi(t) = \sum_{\ell \geq 0} q_\lambda^\ell \left(\prod_{j=1}^{\ell} m_{t+j} \right) p_\phi(t+\ell).}$$

Thus the advantage-level quantity used in the main theory is u_ϕ when the attacker operates at the TD level. A local TD payload $p_\phi(t) = f_\phi(s_t, a_t)$ generally produces a temporally spread advantage perturbation; it is not merely a rescaled copy of f_ϕ.

For a fixed rollout, a desired advantage-level payload $f_\phi(t)$ can be realized exactly through a backward TD payload

$$\boxed{p_\phi^{\mathrm{match}}(t) = f_\phi(t) - q_\lambda m_{t+1} f_\phi(t+1).}$$

With $f_\phi(T) = 0$ at the rollout boundary, substitution into the recursion gives $u_\phi(t) = f_\phi(t)$ for every timestep. This exact construction requires the TD/GAE component to see the completed rollout; it is therefore not assumed in the main threat model. It is useful for a unit-level equivalence check if a TD-level implementation is added later.

For example, upper TD-error clipping corresponds to

$$p_+(t) = \min\{\delta_t, c_+\} - \delta_t = -(\delta_t - c_+)_+,$$

and lower clipping corresponds to

$$p_-(t) = \max\{\delta_t, -c_-\} - \delta_t = (-\delta_t - c_-)_+.$$

Their induced advantage perturbations are $u_\pm = \mathrm{GAE}(p_\pm)$, which are the $-B_+$ and $+B_-$ quantities analyzed in the main theory.

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